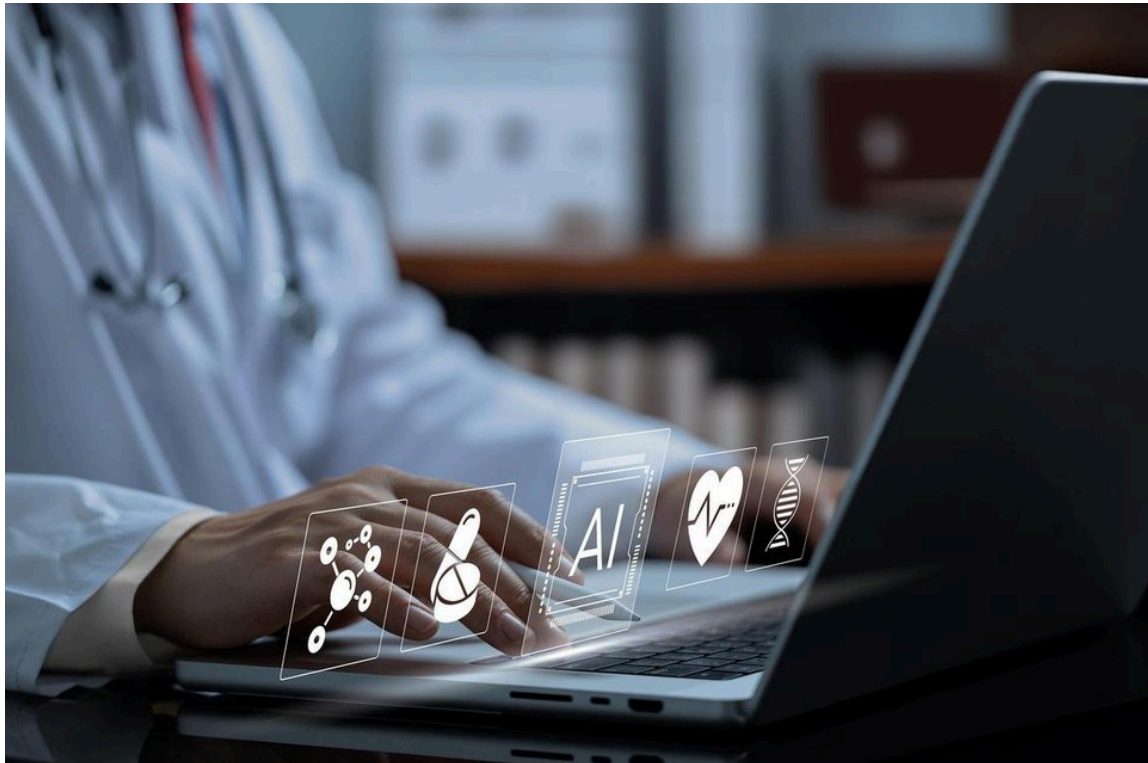


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Predictive AI can be the prescription for future of Singapore's healthcare

The real transformation is not about replacing doctors with AI, but about empowering them with it.

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Artificial intelligence will have a profound impact on doctors' administrative burden, says the writer.

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Ilya Burkov

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Singapore's Health Minister Ong Ye Kung's announcement in June 2025 that the country **will integrate predictive artificial intelligence**

(AI) into its national Healthier SG marks a significant turning point.

Leveraging health records and AI to forecast disease risk years in advance will allow the industry to chart a course for a more proactive, preventive and personalised era of medicine.

Yet, this news might spark unease, especially around AI outperforming humans in the medical field, from simple administrative tasks to complex diagnostic responsibilities such as treating diseases or even mapping tumours.

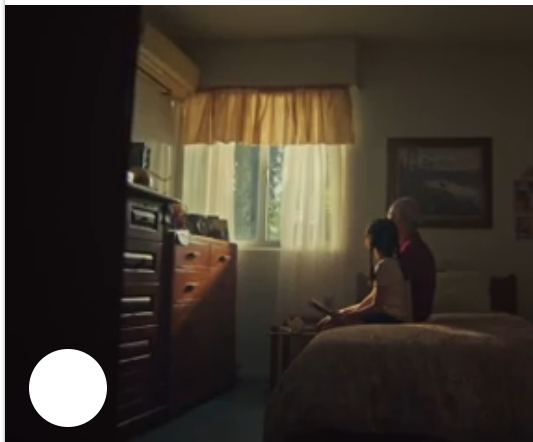
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It is easy to imagine a future where the physician's nuanced judgment is replaced by the logic of an algorithm. However, this vision is misguided. The real transformation in healthcare is not about replacing doctors with AI, but about empowering them with it.

The central challenge of our time is not adapting to a world run by AI, but in preparing the next generation of clinicians to wield these powerful new tools.

The diagnostic co-pilot

A modern-day clinician faces a deluge of data on a daily basis. For example, a radiologist might review dozens of complex scans every hour, where fatigue can potentially lead to errors.

Now, imagine an AI co-pilot trained on millions of images.

As demonstrated in a recent study from Northwestern Medicine, such a tool can match the accuracy of doctors in outlining lung tumours on CT scans, and can also flag high-risk areas that are sometimes missed.

When data analysis is powered by AI, the radiologist can focus on the most complex aspects of the case, consult with colleagues, and, most importantly, communicate with the patient. The technology cuts through vast datasets by providing unbiased second opinions that reduce diagnostic errors and standardise care quality. This augmentation is critical in pattern recognition and, by extension, the delivery of care.

Some might view such advanced capabilities with apprehension, but the reality is quite the opposite. In fact, studies have shown that AI systems can increase accuracy in complex cases, to the pleasant surprise of experts. This isn't a threat to the profession; it's an opportunity to leverage it.

For instance, predictive AI is already advancing patient care at Singapore General Hospital, which has deployed CARES-ML, a predictive AI tool achieving over 90 per cent accuracy in assessing surgery risk using patient data.

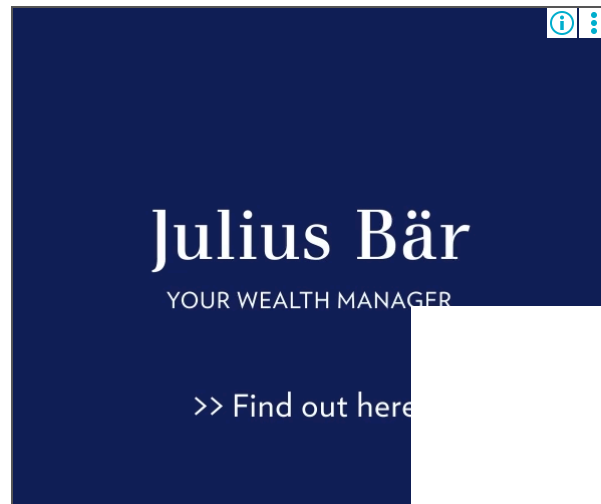
By shouldering the burden of sifting through immense datasets, AI provides clinicians with a clearer signal from the noise, enabling faster, more confident, and ultimately more accurate decisions.

From data tsunami to targeted therapies

This collaborative power extends deep into the laboratory, where the pace of discovery has long been bottlenecked by the sheer volume of biological data.

The journey from identifying a potential new drug to bringing it to market is a decade-long, multibillion-dollar marathon. Researchers traditionally had to screen thousands of molecular compounds, a process heavily constrained by time, resources and clunky infrastructure.

However, AI is rapidly transforming this landscape. By creating "digital twins" of cells and molecules, scientists can now simulate how a drug might behave in the body, often before it ever reaches a clinical trial.



For instance, Tel Aviv-based startup Converge Bio used an AI platform to analyse 36 million individual cells to better understand disease mechanisms, compressing a months-long task into a matter of days. This allows researchers to “fail faster” in a virtual environment, quickly ruling out unpromising drug candidates and focusing resources on those with the highest chance of success.

Similarly, the National Heart Centre Singapore is leveraging AI for rapid coronary artery disease prediction by interpreting heart image scans, significantly reducing analysis time from hours to minutes.

This acceleration is fundamental to realising the promise of precision medicine, a cornerstone of Singapore’s national strategy. To tailor a cancer treatment to a specific tumour’s genetic make-up, we must first make sense of that patient’s unique genomic data.

AI provides the computational horsepower to do just that, turning a torrent of information into actionable clinical insights.

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Mitigating risks to advance the human touch

Perhaps the most profound impact of AI will be in tackling a far more mundane, yet pervasive, problem: the administrative burden that consumes a doctor's day.

For every hour of patient interaction, physicians often spend two hours on paperwork and updating electronic health records.

Here, tools like AI scribes are already making a tangible difference.

By transcribing patient consultations in real time, they save clinicians hours of administrative work, allowing them to be more present and engaged during appointments.

This is the very “human touch” that is irreplaceable. AI cannot empathise, but it can create the time and space for a human to do so.

AI's deployment is unlocking exciting possibilities across fields like radiology, where its efficiency can streamline tasks, paving the way for professionals to embrace evolving roles and explore innovative new opportunities.

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But beyond the workforce implications, AI implementation introduces some significant challenges, including cost.

Substantial upfront investments are required for advanced AI infrastructure. There are also critical concerns regarding algorithmic biases, as AI trained on unreliable data can lead to skewed decisions disproportionately affecting certain patient populations. The

inherent potential for AI models to make errors also underscores the vital need for vigilant oversight.

To navigate these challenges, robust governance is needed to protect patient data, ensure algorithmic fairness and prevent errors, while keeping the human clinician firmly in the loop as a pilot, not a passenger.

The solution is not to halt progress, but to build frameworks that ensure transparency, address bias proactively and establish clear lines of responsibility. This includes ensuring that high-performance AI compute becomes more accessible and cost-efficient for all players in the healthcare ecosystem, from large institutions to smaller start-ups.

We must invest not only in the algorithms, but in the scalable cloud infrastructure that underpins them and, most critically, in the training of our healthcare professionals.

Clinics of the future will not be staffed by robots. They will be led by clinicians who, empowered by AI, are able to deliver a standard of care more precise, personal and profoundly human than ever before.

- Dr Ilya Burkov leads global healthcare and life sciences growth at Nebius, where he helps biotech teams leverage AI to accelerate drug discovery. He previously worked in several healthcare start-ups across the UK and Europe.

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